One repository, every lab's cells.

A habit for scientists, a certificate for buyers, and a place where cell data converts.

← all documents · Kat Titterton · San Francisco

Vision

Every lab that images cells should know where its cells sit among everyone else's, and every dataset made from cells should carry that answer with it. Datum Line is one shared repository of raw cell images from every lab, on one map. It starts as the map, grows into a certificate, then subscriptions: scientists, AI agents and companies pay for access to images that are comparable, certified and traceable to the lab that made them. The arrow after subscriptions is left open on purpose.

The habit comes first, the way Strava's segments place a run among everyone who ran the same stretch of road. Here the road is the cell line, and the field is every lab that ran it. Everything below serves one rule: **uploading must be effortless, and the answer must be worth looking at.**

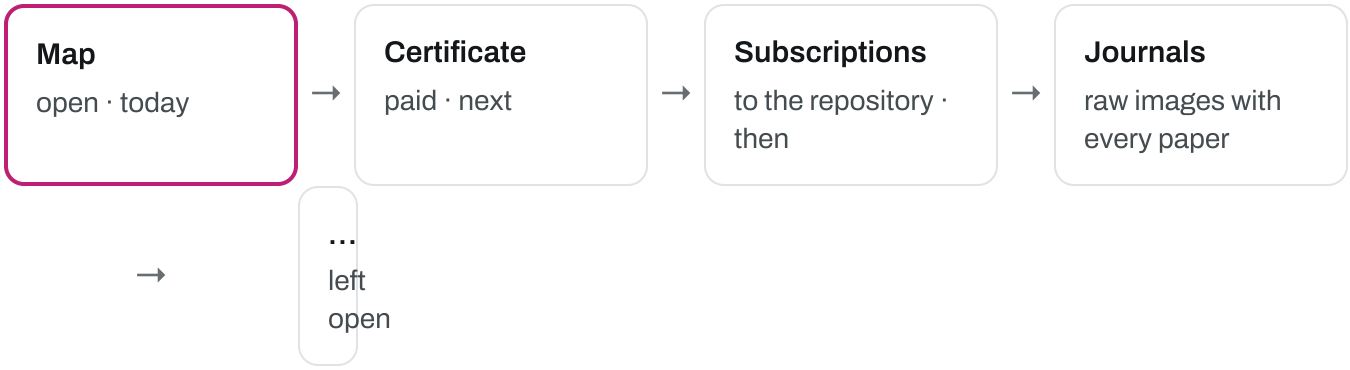
01 Why now

Cell data has a price. One vendor pays its experts more than $1.5M a day, about $85 an hour on average, at a $10B valuation. [10] Roche and Genentech paid Recursion a second $30M milestone for an accepted whole-genome cell map; $213M from the partnership to date. [11] A $1.8B virtual-cell data initiative, part new money and part existing funding; Meta, Google DeepMind and Isomorphic put in $300M together and get a year of exclusive access first. [12]

Cell data does not convert. EU-OPENSCREEN ran one Cell Painting protocol at 4 sites, with cells, serum and dye lots supplied centrally. Within a site, 84 to 94% of compounds replicate; across sites, 54 to 68% pair, and the authors call the profiles "not directly comparable between imaging sites" [13]. On the public JUMP data, a classifier names the lab from DMSO wells alone at 99.9% against 9.1% chance. Whoever makes cell data comparable sets its exchange rate.

02 The path

Fig. 1 **A starting point, not an end state**



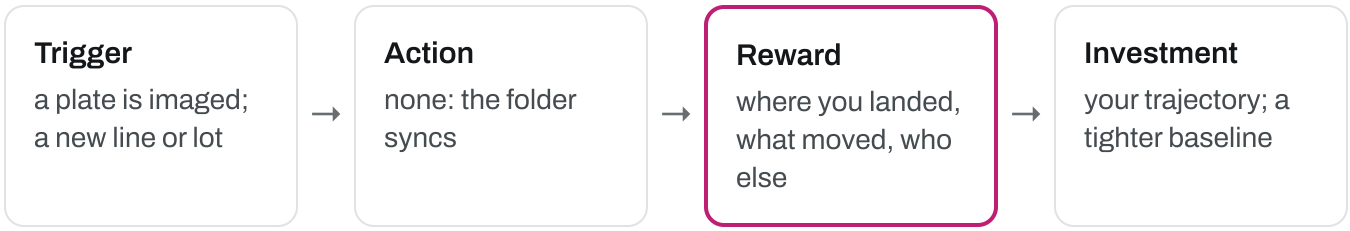
each stage pays for the next, and none is promised a date

STAGE	WHAT IT IS	WHO PAYS	WHAT HAS TO BE TRUE	STATUS
Map	every lab's DMSO wells placed on one map; the box, the code and the public reference	nobody: open, and never crippled	labs upload the controls they already image	today: 11 labs, 129 batches, 1,871 plates of public data
Certificate	where a dataset sits, who made it and what moved, attached to the data	companies that buy cell data, per dataset; groups that screen, per line per quarter	the map holds for the line	next: prices proposed on board A15
Subscriptions	certified datasets, listed by the labs that made them	subscribers, for access to data from many labs, not ownership; each lab gets a share	buyers trust the certificate; labs choose to list	then

STAGE	WHAT IT IS	WHO PAYS	WHAT HAS TO BE TRUE	STATUS
…	what the map makes possible next: journals asking for raw control images with every paper, alerts when a lot drifts across labs, a graded benchmark, models trained on every lab's controls			open

03 The habit loop

Fig. 2 **Every upload makes the next answer worth opening**



↺ back to Trigger: the reference grows, the next reading is sharper

STEP	IN DATUM LINE	DESIGN RULE
Trigger	the microscope finishes a plate; a new line or lot arrives	the trigger is the instrument, not an email
Action	nothing: the export folder syncs	any click is a leak; the reference starves on friction
Reward	where you landed, which features moved, who else moved	real uncertainty: nobody knows their position until they look
Investment	your history becomes a trajectory; your baseline tightens	leaving costs your own past, which is honest switching cost

04 Six magic moments

Each moment turns an image the lab already has into something it could not see before, and each is ordered by when a new customer meets it.

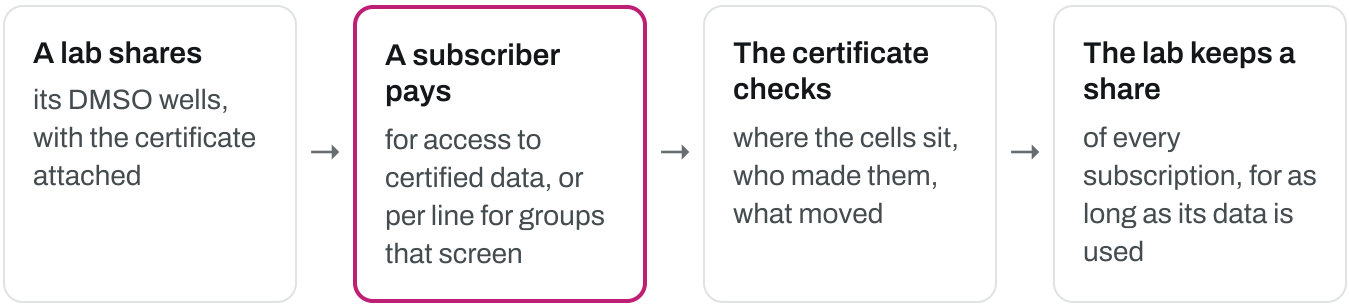
#	MOMENT	WHAT THE LAB SEES	WHEN
1	**Your past, diagnosed**	old plates dragged in on day one appear as a path across the field map: "you left the envelope in March"	first ten minutes
2	**Show the cells, not the score**	when a plate moves, its nuclei next to their nearest neighbours from labs in distribution: same size, more texture	every flag
3	**The fix**	what the in-distribution labs do differently: seeding density, media or serum lot	after a flag
4	**A forecast**	"at your current drift, this line leaves the envelope in about five weeks"	after three readings
5	**Someone else's plate saves your screen**	a lot you use is drifting at other labs, before your own plates show it	once labs share lots
6	**The certificate as currency**	a verifiable link for a methods section, a CRO handoff, a grant report	when results leave the lab

Moments 1 to 4 work for a single lab on day one, against the public reference. Moments 5 and 6 exist only once the network does, and no competitor can copy them without the labs.

05 Subscriptions

A certificate makes a dataset legible to someone who did not make it. The repository is where it is shared: subscribers get access for a period, not ownership.

Fig. 3 **Every subscription pays the labs that made the data**



↺ every listing widens the map, and a wider map makes every certificate sharper

WHAT SUBSCRIBERS GET	WHY IT IS WORTH PAYING FOR
Certified datasets	they convert: on the map, agreement between labs on the same drugs goes from 0.75 to 0.78 (board A14)
Early access	a window before data goes public, which consortia already sell: a year, at the virtual-cell initiative [12]
The benchmark	AI labs and model builders graded on planted faults, with a cell-count baseline on every task (board A16)

What never gets sold. A lab's treated data, its results, its name without consent, a ranking of instruments, or a verdict for hire. The certificate can be bought; the answer cannot. Prices and shares are proposals (boards A15 and A22) until the first partner labs and the first buyer set them.

06 Trust, without the token

The open-science half of the product, and the reason a lab would share.

PROMISE	HOW	STATUS
Control wells only	nothing else leaves a lab: no compound, no target, no result; the frame is applied on the lab's side	today

PROMISE	HOW	STATUS
Every lab leaves a fingerprint	99.9% of DMSO wells name their own lab on plates the model never saw, so a dataset claimed from one lab and made in another is caught	today
Signed and timestamped	every certificate signed, its fingerprint written to a public log: anyone can check it, nobody can backdate it; a blockchain is one such log, not a requirement	proposed
The lab keeps its data	listing is a choice and can be withdrawn; a share of every subscription goes to the lab that made the data	proposed
Known subscribers	subscribers verified at sign-up; unusual access flagged; every access in a log the members can read	proposed
Low risk by design	images of cell lines: no human genetic data and no pathogen data; the map holds DMSO wells only	today

For a decentralized-science audience this is the point: open data, verifiable records and contributor upside, without a token to speculate on and without betting on anyone's results.

07 The game

Every batch a lab sends is ranked against every batch the field has seen, on its DMSO wells alone: how typical the cells are, how alike the plates are, how steady the lab is from batch to batch, and cells per well against the field. Nothing in the score touches a treated well or a result, so the only way to climb is better culture and imaging practice. On the public data, lab 5 holds the field's longest streak, 33 batches (results/e14). A lab sees its rank, its personal best, its streak and the lab to beat.

DO	DON'T	WHY
rank practice, never results	rank a lab's results or its drugs	the only way up is better culture and imaging
labs are numbers until they choose a name	publish a lab's name or rank without consent	shame makes labs hide their worst plates, which teach the field most

DO	DON'T	WHY
credit for uploading flagged plates	penalise a batch for sitting off the map	a failure uploaded is worth more than a success
explain every flag with cells and a likely fix	show a bare red dot	a verdict without a reason reads as judgement
let the instrument trigger uploads	send nudges to protect a streak	the reward is the answer, not the visit

08 Metrics

North star: the share of a lab's plates that upload with no human action. If it is high, the loop turns; if people have to remember, nothing else matters.

METRIC	WHAT IT TELLS YOU
automatic upload share	the habit exists without effort
flagged plates uploaded, not deleted	labs trust the product with bad news
labs per line, lines per lab	the map is widening
datasets listed with a certificate	labs trust the repository with their data
renewing subscribers	access is worth paying for, again
share of each subscription paid to the lab	contributors are earning, so more will list
certificates cited outside the lab	the certificate has become currency

09 What could go wrong

RISK	WHY IT IS REAL	ANSWER
cold start	no map without labs, no labs without a map	seed from JUMP's public controls; U2OS first, any other line from its first batch
labs will not list	data is a lab's edge, and pharma data is intellectual property	the certificate is free and useful alone; listing is optional and pays the lab
public data must stay free	NIH-funded labs must share the data behind their findings	charge for certification, comparability and compute, never for access to data a lab owes the public
buyers want designed data at scale	the deals above buy factories and consortia, not leftovers	certify factory and consortium data too; the benchmark grades their models
a platform builds its own map	phenomics companies hold the most images	nobody can certify their own drift; the reference has to be neutral and open

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